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Chapter 1

Linear second order equations

A linear ordinary differential equation (ODE) of order 2 has the form

$$y'' + p(x)y' + q(x)y = r(x) \tag{1.1}$$

where $p(x)$, $q(x)$ and $r(x)$ are functions of x. The functions $p(x)$, $q(x)$ and $r(x)$ are called the coefficients of the equation. When $p(x)$, $q(x)$ and $r(x)$ are constants we say that the equation has constant coefficients, otherwise it has variable coefficients. If $r(x) = 0$ then (1.1) has the form

$$y'' + p(x)y' + q(x)y = 0 \tag{1.2}$$

and is then called **homogeneous** to indicate that all terms are of the first degree in y and its derivatives. The original equation (1.1) is called **non-homogeneous** because it contains a term $r(x)$ which dos not depend on y.

Example 1. The equation

$$(1 - x)y'' + 3xy' + 6y = 0$$

is homogeneous second order linear differential equation while the equation

$$y'' + 3y' + 9y = e^{\cos x}$$

is a nonhomogeneous second order linear differential equation.

In this course we will consider two cases, homogeneous and non-homogeneous differential equations.

1.1 Linear combinations of solutions

Theorem 1.1. *Let y_1 and y_2 be solutions to the homogeneous equation*

$$y'' + py' + qy = 0 \tag{1.3}$$

Then any linear combination $c_1y_1 + c_2y_2$ of y_1 and y_2 where c_1 and c_2 are constants, is also a solution to (1.3)

Example 2. Given that $y_1(x) = e^{2x}\cos 3x$ and $y_2(x) = e^{2x}\sin 3x$ are solutions to the homogeneous equation $y'' - 4y' + 13y = 0$. Find a solution to this equation that satisfies the initial conditions $y(0) = 5$ and $y'(0) = -2$.

Solution. *As a consequence of Theorem 1.3, any linear combination*

$$y(x) = c_1e^{2x}\cos 3x + c_2e^{2x}\sin 3x \tag{1.4}$$

with c_1 and c_2 arbitrary constants, will be a solution to $y'' - 4y' + 13y = 0$.

Differentiating,

$$y'(x) = c_1\big(2e^{2x}\cos 3x - 3e^{2x}\sin 3x\big) + c_2\big(2e^{2x}\sin 3x + 3e^{2x}\cos 3x\big) \tag{1.5}$$

Substitute (1.4) and (1.5) into initial conditions yields $c_1 = 5, 2c_1 + 3c_2 = -2$ which implies $c_2 = -4$. Therefore the solution that satisfies the initial conditions is $y(x) = 5e^{2x}\cos 3x - 4e^{2x}\sin 3x$.

1.2 Linear dependence of functions

Definition 1. Two functions y_1 and y_2 are said to be *linearly dependent* on an interval I if there exist nonzero constants c_1 and c_2 such that

$$c_1y_1(x) + c_2y_2(x) = 0$$

for all x in I. If two functions are not linearly dependent, they are said to be *linearly independent*.

Example 3. Determine whether the following pairs of functions y_1 and y_2 are linearly dependent or independent:

(a) $y_1(x) = e^{3x}, \quad y_2(x) = x + 1$

(b) $y_1(x) = \sin 2x, \quad y_2(x) = \cos x \sin x$

Solution. — *Exercise* —

1.3 Representation of solution (homogeneous case)

Theorem 1.2. *Let y_1 and y_2 denote two solutions on a subinterval (a, b) of*

$$y'' + p(x)y' + q(x)y = 0 \tag{1.6}$$

where p and q are continuous on (a, b). Suppose at some point x_0 in (a, b) these solutions satisfy

$$y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \neq 0. \tag{1.7}$$

Then every solution of (1.6) on (a, b) can be expressed in the form

$$y(x) = c_1y_1(x) + c_2y_2(x)$$

where c_1 and c_2 are constants.

Because the expression on the left-hand side of (1.7) plays an important role in the theory, we adopt the following terminology.

1.4 Wronskian

Definition 2. For any two differentiable functions y_1 and y_2, the function

$$\begin{aligned} W[y_1, y_2](x) &= y_1(x)y_2'(x) - y_1'(x)y_2(x) \\ &= \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} \end{aligned} \tag{1.8}$$

is called the Wronskian of y_1 and y_2.

1.5 Fundamental solution set

A pair of solutions $\{y_1, y_2\}$ of $y''+py'+qy = 0$ on (a, b) is called a fundamental solution set if $W[y_1, y_2](x_0) \neq 0$ for all x_0 in (a, b).

Example 4. Let $y_1 = e^x$ and $y_2 = e^{-x}$. Then

$$\begin{aligned} W[y_1, y_2](x) &= y_1(x)y_2'(x) - y_1'(x)y_2(x) \\ &= \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} \\ &= e^x(-e^{-x}) - e^x(-e^{-x}) \\ &= -2 \neq 0. \end{aligned}$$

Therefore $\{e^x, e^{-x}\}$ is a fundamental solution set.

1.6 Procedure for solving homogeneous equation

To determine all solutions to $y'' + py' + qy = 0$,

(a) find two solutions y_1, y_2 that constitute a fundamental solution set

(b) form the linear combination

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where c_1 and c_2 are arbitrary constants, to obtain a general solution.

Example 5. Given that $y_1(x) = \cos 3x$ and $y_2(x) = \sin 3x$ are solutions to

$$y'' + 9y = 0 \tag{1.9}$$

on $(-\infty, \infty)$, find a general solution to (1.9).

Solution. *First we verify that* $\{\cos 3x, \sin 3x\}$ *is a fundamental solution set.*

$$\begin{aligned} W[y_1, y_2](x) &= \begin{vmatrix} \cos 3x & \sin 3x \\ -3\sin 3x & 3\cos 3x \end{vmatrix} \\ &= (\cos 3x)(3\cos 3x) - (-3\sin 3x)(\sin 3x) \\ &= 3\cos^2 3x + 3\sin^2 3x \\ &= 3(\cos^2 3x + \sin^2 3x) \\ &= 3(1) \\ &= 3 \neq 0. \end{aligned}$$

Thus $\{\cos 3x, \sin 3x\}$ *forms a fundamental solution set, and a general solution to* (1.9) *is*

$$y(x) = c_1 \cos 3x + c_2 \sin 3x.$$

Theorem 1.3. *Suppose* y_1, y_2 *are solutions to* $y'' + p(x)y' + q(x)y = 0$. *The solutions* y_1, y_2 *are linearly dependent if and only if* $W[y_1, y_2] = 0$. *Equivalently* y_1, y_2 *are linearly independent if and only if* $W[y_1, y_2] \neq 0$.

1.7 Fundamental sets, linear independence and the Wronskian

If y_1, y_2 are solutions to $y'' + p(x)y' + q(x)y = 0$ on (a, b), then the following statements are equivalent:

(i) $\{y_1, y_2\}$ is a fundamental solution set on (a, b).

(ii) y_1 and y_2 are linearly independent on (a, b).

(iii) The Wronskian $W[y_1, y_2](x)$ is never zero on (a, b).

Example 6. Show that $y_1(x) = x^{-1}$ and $y_2(x) = x^3$ are solutions to

$$x^2y'' - xy' - 3y = 0 \tag{1.10}$$

on the interval $(0, \infty)$ and give a general solution.

Solution. *Substitute* y_1, y_1' *and* y_1'' (1.10)

$$\begin{aligned} x^2(2x^{-3}) - x(-x^{-2}) - 3(x^{-1}) &= 2x^{-1} + x^{-1} - 3x^{-1} \\ &= 0, \end{aligned}$$

and likewise for y_2,

$$\begin{aligned} x^2(6x) - x(3x^2) - 3(x^3) &= 6x^3 - 3x^3 - 3x^3 \\ &= 0. \end{aligned}$$

The solution function x^{-1} and x^3 are linearly independent on $(0, \infty)$ [neither is a constant multiple of the other on $(0, \infty)$].

$$\begin{aligned} W[y_1, y_2](x) &= \begin{vmatrix} x^{-1} & x^3 \\ -x^{-2} & 3x^2 \end{vmatrix} \\ &= 3x^1 + x^1 \\ &= 4x \neq 0. \end{aligned}$$

Therefore the general solution is $y(x) = c_1x^{-1} + c_2x^3$.

1.8 Existence and uniqueness of solution

Theorem 1.4. *Suppose* $p(x), q(x)$ *and* $g(x)$ *are continuous on an interval* (a, b) *that contains the point* x_0. *Then, for any choice of the initial values* y_0 *and* y_1, *there exists a unique solution* $y(x)$ *on the whole interval* (a, b) *to the initial value problem*

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = g(x), \quad y(x_0) = y_0,\ y'(x_0) = y_1$$

Chapter 2

Homogeneous linear equations with constant coefficients

Let's consider the homogeneous linear second order differential equation with constant coefficients

$$ay'' + by' + cy = 0 \tag{2.1}$$

where a ($\neq 0$), b, and c are real constants. Because constant functions are everywhere continuous, Theorem 1.4 guarantees that equation (2.1) has solutions defined for all x in $(\infty, -\infty)$. If we can find two linearly independent solution to equation (2.1), say y_1 and y_2, then we can express a general solution in the form

$$y = c_1 y_1 + c_2 y_2, \tag{2.2}$$

where c_1 and c_2 are arbitrary constants.

A look at equation (2.1) tells us that a solution of (2.1) must have property that its second derivative is expressible as a linear combination of its first and zero-th derivatives. This suggests that we try to find a solution of the form $y = e^{rx}$, since derivatives of e^{rx} are just constants times e^{rx}. If we substitute $y = e^{rx}$ into (2.1), we obtain

$$\begin{aligned} ar^2 e^{rx} + bre^{rx} + ce^{rx} &= 0 \\ e^{rx}\big(ar^2 + br + c\big) &= 0. \end{aligned}$$

Since $e^{rx} \neq 0$ this implies

$$ar^2 + br + c = 0.$$

Consequently, $y = e^{rx}$ is a solution to $ay'' + by' + cy = 0$ if and only if r satisfies $ar^2 + br + c = 0$. The equation $ar^2 + br + c = 0$ is called the **auxiliary equation**. Since the auxiliary equation is quadratic, its roots are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

2.1 Distinct real roots

When the discriminant, $b^2 - 4ac$, is positive ($b^2 - 4ac > 0$), the roots r_1 and r_2 are real and distinct.

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

It can be shown that $W(e^{r_1 x}, e^{r_2 x}) \neq 0$. Therefore $e^{r_1 x}$ and $e^{r_2 x}$ are independent solutions to $ay'' + by' + cy = 0$.

Then the general solution is $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$ where c_1 and c_2 are arbitrary constants.

2.2 Repeated Roots

If the auxiliary equation has a repeated root r, then two linearly independent solutions to $ay'' + by + cy = 0$ are e^{rx} and xe^{rx} and a general solution is

$$y = c_1 e^{rx} + c_2 x e^{rx}. \tag{2.3}$$

The roots for the auxiliary equations are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Since the roots are repeated, $b^2 - 4ac = 0$. This implies $r_1 = r_2 = -\frac{b}{2a}$. A solution is $y_1 = e^{-\frac{b}{2a}x}$.

To find the second solution which is linearly independent to the first solution, we use the variation of parameter method.[1]

Let $y = v(x)y_1$. Then

$$\begin{aligned} y' &= v'y_1 + vy_1' \\ y'' &= v''y_1 + v'y_1' + v'y_1' + vy_1'' \\ &= v''y_1 + 2v'y_1' + vy_1'' \end{aligned}$$

Substitute y, y' and y'' into

$$ay'' + by' + cy = 0 \tag{2.4}$$

to obtain

$$a(v''y_1 + 2v'y_1' + vy_1'') + b(v'y_1 + vy_1') + cvy_1 = 0$$

[1] see later chapter

and rearranging,

$$(ay_1)v'' + (2ay_1' + by_1)v' + (ay_1'' + by_1' + cy_1)v = 0.$$

Now by (2.4), $ay_1'' + by_1' + cy_1 = 0$.
We want $2ay_1' + by_1 = 0$ which implies $\frac{1}{y_1}dy = -\frac{b}{2a}dx$. Solve to obtain

$$y_1 = e^{-\frac{b}{2a}x}.$$

Since $a \neq 0$ and $y_1 \neq 0$, we have $v''(x) = 0$ and therefore

$$\begin{aligned} v'(x) &= c_1, \\ v(x) &= c_1 x + c_2 \end{aligned} \tag{2.5}$$

for some constants c_1 and c_2.
Substitute (2.5) into $y = v(x)y_1$, to get

$$\begin{aligned} y &= (c_1 x + c_2)y_1 \\ &= c_1 x y_1 + c_2 y_1 \\ &= c_1 x e^{-\frac{b}{2a}x} + c_2 e^{-\frac{b}{2a}x}. \end{aligned}$$

2.3 Complex Roots

When the discriminant, $b^2 - 4ac$, is negative ($b^2 - 4ac < 0$), the roots of the auxiliary equation, r_1 and r_2 are complex.

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

If $\sqrt{b^2 - 4ac} < 0$,

$$\begin{aligned} r_1 &= \frac{-b + \sqrt{-1}\sqrt{4ac - b^2}}{2a} \\ &= \frac{-b}{2a} + i\left[\frac{\sqrt{4ac - b^2}}{2a}\right], \end{aligned} \qquad \begin{aligned} r_2 &= \frac{-b - \sqrt{-1}\sqrt{4ac - b^2}}{2a} \\ &= \frac{-b}{2a} - i\left[\frac{\sqrt{4ac - b^2}}{2a}\right]. \end{aligned}$$

Let $\alpha = \frac{-b}{2a}$ and $\beta = \frac{\sqrt{4ac-b^2}}{2a}$. Then $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$.

By Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$, we can express $y(x)$ as

$$\begin{aligned}
y(x) &= c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x} \\
&= e^{\alpha x}\big[c_1 e^{i\beta x} + c_2 e^{-i\beta x}\big] \\
&= e^{\alpha x}\Big\{c_1\big[\cos\beta x + i\sin\beta x\big] + c_2\big[\cos(-\beta x) + i\sin(-\beta x)\big]\Big\} \\
&= e^{\alpha x}\Big\{c_1\big[\cos\beta x + i\sin\beta x\big] + c_2\big[\cos\beta x - i\sin\beta x\big]\Big\} \\
&= e^{\alpha x}\big[(c_1 + c_2)\cos\beta x + (c_1 - c_2)i\sin\beta x\big] \\
&= e^{\alpha x}\big[k_1\cos\beta x + k_2\sin\beta x\big] \quad \text{where } k_1 = c_1 + c_2, k_2 = (c_1 - c_2)i \\
&= k_1 \underbrace{e^{\alpha x}\cos\beta x}_{y_1} + k_2 \underbrace{e^{\alpha x}\sin\beta x}_{y_2}
\end{aligned}$$

Example 7. Find a general solution for the equations

(a) $6y'' - 11y' + 3y = 0$

(b) $2y'' - 3y' = 0$

(c) $4y'' + 4y' + y = 0$

(d) (c), with initial conditions $y(2) = 1, y'(2) = ?$

(e) $16z'' - 56z' + 49z = 0$

(f) $y'' + 2y' + 4y = 0$

Solution.

(a) The auxiliary equation is $6r^2 - 11r + 3 = (3r - 1)(2r - 3) = 0$. *The roots are* $r_1 = \frac{1}{3}$ *and* $r_2 = \frac{3}{2}$ *which are distinct. Therefore* $e^{\frac{1}{3}x}$ *and* $e^{\frac{3}{2}x}$ *are linearly independent solutions.*

Hence the general solution is $y(x) = c_1 e^{\frac{1}{3}x} + c_2 e^{\frac{3}{2}x}$.

(b) The auxiliary equation is $2r^2 - 3r = r(2r - 3) = 0$. *The roots are* $r_1 = 0$ *and* $r_2 = \frac{3}{2}$ *which are distinct. Therefore* $e^{0x} = 1$ *and* $e^{\frac{3}{2}x}$ *are linearly independent solutions.*

Hence the general solution is $y(x) = c_1 \cdot 1 + c_2 e^{\frac{3}{2}x} = c_1 + c_2 e^{\frac{3}{2}x}$.

2.4 Higher order differential equations

The method of solving homogeneous linear second order differrential equation and constant coefficient can be used to solve any order of differential equation. In general, linear homogeneous differential equation of n order have n independent solutions.

Example 8. Find a general solution to $y''' + 3y'' - y' - 3y = 0$.

Solution. *Auxilliary equation is* $r^3 + 3r^2 - r - 3 = (r-1)(r+1)(r+3) = 0$ *so* $r_1 = 1, r_2 = -1$ *and* $r_3 = -3$.

Therefore the general solution is $y(x) = c_1e^x + c_2e^{-x} + c_3e^{-3x}$.

Example 9. Find a general solution to $y^{(4)} - y''' - 3y'' + 5y' - 2y = 0$.

Solution. *Auxilliary equation is* $r^4 - r^3 - 3r^2 + 5r - 2 = 0$ *which implies* $r_1 = 1, r_2 = 1, r_3 = 1$ *and* $r_4 = -2$.

The general solution is $y(x) = c_1e^x + c_2xe^x + c_3x^2e^x + c_4e^{-2x}$.

Chapter 3

Superposition and non-homogeneous equations

The differential operator L defined by

$$L[y](x) = y'' + p(x)y' + q(x)y \tag{3.1}$$

can be viewed as a "black box" with input the function $y(x)$ and output the function on the right-hand side of (3.1). Suppose the input functions $y_1(x)$ and $y_2(x)$ yield, respectively, the output functions $g_1(x)$ and $g_2(x)$, that is

$$L[y_1] = g_1(x), \quad L[y_2] = g_2(x).$$

Then since L is a linear operator, an input consisting of the linear combination $c_1y_1(x) + c_2y_2(x)$ produces the output $c_1g_1(x) + c_2g_2(x)$ in the same combination.

3.1 Superposition Principle

Theorem 3.1. *Let y_1 be a solution of the differential equation*

$$L[y_1] = g_1(x)$$

and let y_2 be a solution of

$$L[y_2] = g_2(x)$$

where L is a linear differential operator. Then for any constants c_1 and c_2, the function $c_1y_1 + c_2y_2$ is a solution to the differential equation

$$L[y] = c_1g_1(x) + c_2g_2(x).$$

Example 10. Given that $y_1(x) = -\frac{x}{3} - \frac{2}{9}$ is a solution to

$$y'' + 2y' - 3y = x,$$

and $y_2(x) = -\frac{e^{2x}}{5}$ is a solution to

$$y'' + 2y' - 3y = e^{2x},$$

find a solution to

$$y'' + 2y' - 3y = 4x - 5e^{2x}.$$

Solution. *Let* $L[y] = y'' + 2y' - 3y$. *We are given that*

$$L[y_1] = g_1(x) = x \quad and L[y_2] = g_2(x) = e^{2x}.$$

Since $4x - 5e^{2x} = 4g_1(x) - 5g_2(x)$, *then, by the superposition principle,*

$$c_1y_1 + c_2y_2 = 4\left(-\frac{x}{3} - \frac{2}{9}\right) - 5\left(\frac{e^{2x}}{5}\right) \qquad = -\frac{4x}{3} - \frac{8}{9} - e^{2x}$$

is a solution to $y'' + 2y' - 3y = 4x - 5e^{2x}$.

Theorem 3.2. *Let* $y_p(x)$ *be a particular solution to the non-homogeneous equation*

$$y'' + p(x)y' + q(x)y = g(x) \tag{3.2}$$

on the interval (a, b) *and let* $y_1(x), y_2(x)$ *be linearly independent solutions on* (a, b) *of the corresponding homogeneous equation*

$$y'' + p(x)y' + q(x)y = 0.$$

Then every solution of (3.2) *on the interval* (a, b) *can be expressed in the*

$$y(x) = y_p(x) + c_1y_1(x) + c_2y_2(x)$$

which is the general solution to (3.2).

3.2 Procedure for solving non-homogeneous equations

To solve $y'' + p(x)y' + qy = g(x)$,

(a) determine the general solution $y_h = c_1y_1 + c_2y_2$ of the corresponding homogeneous equation,

(b) find a particular solution, y_p, of the given non-homogeneous equation,

(c) form the sum of the particular solution and a general to the homogeneous equation; that is,

$$\begin{aligned} y &= y_h + y_p \\ &= c_1 y_1 + c_2 y_2 + y_p \end{aligned}$$

which is the general solution to the given equation.

Example 11. Given that $y_p = x^2$ is a particular solution to $y'' - y = 2 - x^2$, find the general solution to the differential equation.

Solution. *The homogeneous equation is* $y'' - y = 0$ *and the auxiliary equation is* $r^2 - 1 = 0$ *with distinct roots* $r = \pm 1$. *The general solution to the homogeneous equation is*

$$y_h = c_1 e^x + c_2 e^{-x}$$

Therefore the general solution to the nonhomogeneous equation is

$$\begin{aligned} y(x) &= y_h + y_p \\ &= c_1 e^x + c_2 e^{-x} + x^2. \end{aligned}$$

Chapter 4

Method of undetermined coefficients

The method of undetermined coefficients is a simple procedure for finding a particular solution to a non-homogeneous term $g(x)$ where $g(x)$ is a polynomial, exponential function, $\sin x$ or $\cos x$.

Example 12. Find a particular solution to

$$L[y](x) = y'' + 3y' + 2y = 3x + 1.$$

Solution. *We want to find a function $y_p(x)$ such that $L[y_p](x)$ is a linear function of x, namely, $3x + 1$. Now what kind of functions y_p "end up" as linear functions after applications of the operator L? Certainly if y_p linear, so is $L[y_p](x)$. So let's try the form*

$$y_p(x) = Ax + B$$

and attempt to match up $L[y_p](x)$ with $3x + 1$. Since

$$L[y_p](x) = 0 + 3A + 2(Ax + B) = 2Ax + (3A + 2B)$$

we need to solve the equation

$$2Ax + (3A + 2B) = 3x + 1.$$

Two polynomials are equal when their corresponding coefficients are equal, so we set

$$\begin{aligned} 2A &= 3 \\ 3A + 2B &= 1 \end{aligned}$$

Solving this system gives $A = \frac{3}{2}$ *and* $B = -\frac{7}{4}$. *Thus the function*

$$y_p(x) = \tfrac{3}{2}x - \tfrac{7}{4}$$

is a particular solution since $L[y_p](x) = 3x + 1$.

Throughout this section, L *denotes a linear differential operator with constant coefficients. For such operator, Example* (12) *suggests the following method for finding a particular solution to the equation*

$$L[y](x) = p_n(x)$$

where $p_n(x)$ *is a polynomial of degree* n. *Guess a solution of the indefinite form*

$$y_p(x) = A_n x^n + \cdots + A_1 x + A_0$$

and match the coefficients of $L[y_p](x)$ *with those of* $p_n(x)$*:*

$$L[y_p](x) = p_n(x)$$

This procedure involves solving $n + 1$ linear equations in the $n + 1$ unknowns $A_0, \ldots, A_n$ and usually has a solution. The technique is called the **method of undetermined coefficients**.

The same strategy can be applied to other forms for the nonhomogeneous term.

Example 13. Find a particular solution to

$$L[y](x) = y'' + 3y' + 2y = e^{3x}.$$

Solution. *Here we seek a function* y_p *such that* $L[y_p](x) = e^{3x}$. *If we guess* $y_p(x) = Ae^{3x}$ *where* A *is a constant, then*

$$L[y_p](x) = 9Ae^{3x} + 3(3e^{3x}) + 2(Ae^{3x}) = 20Ae^{3x}$$

Setting $20Ae^{3x} = e^{3x}$ *and solving for* A *gives* $A = \frac{1}{20}$.

$$y_p(x) = \frac{e^{3x}}{20}$$

is a particular solution, since $L[y_p](x) = e^{3x}$.

The method of undetermined coefficients will also work for equations of the form

$$L[y](x) = ae^{\alpha x}$$

where a and α are given constants. Here we would guess

$$L[y_p](x) = ae^{\alpha x}$$

and solve $L[y](x) = ae^{\alpha x}$ for the unknown coefficient A.

Example 14. Find a particular solution to

$$L[y](x) = y'' - y' - y = \sin x. \tag{4.1}$$

Solution. *We seek a function y_p that satisfies $L[y_p](x) = \sin x$. Our initial guess might be $y_p(x) = A \sin x$, in which case we get*

$$y_p(x) = -2A \sin x - A \cos x$$

However, since the right-hand side of (4.1) involves only $\sin x$, this choice of solution would force A (and hence y_p itself) to be zero. This will not work, since $y_p(x) = 0$ is obviously not a solution. So to compensate for the $\cos x$ term, let's try

$$y_p(x) = A \cos x + B \sin x$$

which we can verify that

$$L[y_p](x) = (-2A - B) \cos x + (A - 2B) \sin x$$

This implies

$$\begin{aligned} -2A - B &= 0 \\ A - 2B &= 1, \end{aligned}$$

and solve simultaneously to obtain $A = \frac{1}{5}$ and $B = -\frac{2}{5}$.

Hence $y_p = \frac{1}{5} \cos x - \frac{2}{5} \sin x$ is a particular solution.

More generally, for an equation of the form

$$L[y](x) = a \cos \alpha x + b \sin \beta x$$

the method of undetermined coefficients suggests that we guess

$$y_p(x) = A \cos \alpha x + B \sin \beta x$$

and solve $y_p(x) = a \cos x + b \sin x$ for the unknowns A and B.

Example 15. Find a particular solution to

$$L[y](x) = y'' + 4y' = 3 \cos 2x.$$

Solution. *We first guess*

$$y_p(x) = A \cos 2x + B \sin 2x.$$

Then

$$L[y_p](x) = \underbrace{(4A - 4A)}_{=0} \cos 2x + \underbrace{(4B - 4B)}_{=0} \sin 2x = 3\cos 2x.$$

This means that $y_p(x) = A\cos 2x + B\sin 2x$ *is not a particular solution to* $y'' + 4y' = 3\cos 2x$ *since the left hand side is equal to zero. Let's find the solution to the corresponding homogeneous equation,* $y'' + 4y' = 0$.

The auxiliary equation is $r^4 + 4 = 0$ *which implies* $r = \pm 2i$.

Then the general solution to the homogeneous equation is

$$\begin{aligned} y_h &= c_1 e^0 \cos 2x + c_2 e^0 \sin 2x \\ &= c_1 \cos 2x + c_2 \sin 2x. \end{aligned}$$

Now we assume

$$y_p(x) = Ax\cos 2x + Bx\sin 2x.$$

Then

$$L[y_p](x) = -4A\sin 2x + 4B\cos 2x = 3\cos 2x.$$

This implies $A = 0$ *and* $B = \frac{3}{4}$. *Then the particular solution is*

$$y_p(x) = \tfrac{3}{4}x\sin 2x.$$

In the preceding example, the trial choice for y_p did not work because it was a solution to the corresponding homogeneous equation. However, when we replaced y_p by the function $\overline{y_p} = xy_p$, we were able to find the particular solution to the non-homogeneous equation. This procedure for "repairing" the method of undetermined coefficients is generalized as follows.

In any term in the trial expression for y_p is a solution to the corresponding homogeneous equation, then replace y_p by xy_p. If the latter is also a solution, try x^2y_p , then x^3y_p, etc., ultimately employing x^sy_p, where s is the smallest nonnegative integer such that no term in x^sy_p is a solution to the corresponding homogeneous equation.

Chapter 5

Reduction of order method

This method can be used to find a second linearly independent solution of a second order linear homogeneous equation if the first solution is found by inspection or in some other way.

Let

$$y'' + p(x)y' + q(x)y = 0 \tag{5.1}$$

and suppose y_1 be a solution. The second solution is assumed to be $y_2 = u(x)y_1$ for some function $u(x)$. Then

$$\begin{aligned} y_2' &= uy_1' + u'y_1 \\ y_2'' &= uy_1'' + u'y_1' + u'y_1' + u''y_1 \\ &= uy_1'' + 2u'y_1' + u''y_1 \end{aligned}$$

Substitute into (5.1),

$$\begin{aligned} y_2'' + py_2' + qy_2 &= 0 \\ (uy_1'' + 2u'y_1' + u''y_1) + p(uy_1' + u'y_1) + q(uy_1) &= 0 \end{aligned}$$

rearranging the terms

$$u(y_1'' + py_1' + qy_1) + u''y_1 + u'(2y_1' + py_1) = 0 \tag{5.2}$$

Since y_1 is a solution, equation (5.2) reduces to

$$u''y_1 + u'(2y_1' + py_1) = 0.$$

This implies

$$\begin{aligned} u'' &= -u'\left(2\frac{y_1'}{y_1} + p\right) \\ \frac{u''}{u'} &= -p - 2\frac{y_1'}{y_1}. \end{aligned}$$

Integrating

$$\begin{aligned}
\int \frac{u''}{u'} &= \int \left[-p - 2\frac{y_1'}{y_1}\right] \\
\ln u' &= \int -p dx - 2\ln y_1 \\
&= \int -p dx + \ln \frac{1}{y_1^2} \\
u' &= e^{\int -p dx + \ln \frac{1}{y_1^2}} \\
&= e^{\int -p dx}\ e^{\ln \frac{1}{y_1^2}} \\
&= \frac{1}{y_1^2}\ e^{\int -p dx}.
\end{aligned}$$

This implies

$$u(x) = \int \frac{1}{y_1^2}\ e^{\int -p dx}\ dx.$$

The general solution is then

$$\begin{aligned}
y(x) &= c_1 y_1(x) + c_2 y_2(x) \\
&= c_1 y_1(x) + c_2 \left[\frac{1}{y_1^2}\ e^{\int -p dx}\right] \cdot y_1(x)
\end{aligned}$$

where c_1, c_2 are constants.

Example 16. Solve $xy'' + 2y' + xy = 0$, given that $y_1 = \frac{\cos x}{x}$ is a solution.

Solution. *In standard form the equation is* $y'' + \frac{2}{x}y' + y = 0$. *By the reduction of order method, the second solution is* $y_2 = uy_1$ *where*

$$\begin{aligned}
u(x) &= \int \frac{1}{y_1^2} e^{\int -p dx}\ dx \\
&= \int \frac{1}{(\cos x/x)^2} e^{\int -\frac{2}{x} dx}\ dx \\
&= \int \frac{x^2}{\cos^2 x} e^{-2\ln x}\ dx \\
&= \int \frac{x^2}{\cos^2 x} \cdot \frac{1}{x^2}\ dx \\
&= \int \sec^2 x\ dx \\
&= \tan x.
\end{aligned}$$

Therefore the second solution is

$$\begin{aligned} y_2 &= uy_1 \\ &= \tan x \cdot \frac{\cos x}{x} \\ &= \frac{\sin x}{\cos x} \cdot \frac{\cos x}{x} \\ &= \frac{\sin x}{x}, \end{aligned}$$

and so the general solution is

$$\begin{aligned} y &= c_1 y_1 + c_2 y_2 \\ &= c_1 \frac{\cos x}{x} + c_2 \frac{\sin x}{x}. \end{aligned}$$

Example 17. Solve $y'' - \frac{1}{x}y' - 4x^2 y = 0$, given that $y_1 = e^{-x^2}$ is a solution.

Solution. *The equation is already in standard form. The second solution is* $y^2(x) = u(x)e^{-x^2}$ *where*

$$\begin{aligned} u(x) &= \int \frac{1}{y_1^2} e^{\int -p\,dx}\,dx \\ &= \int \frac{1}{(e^{-x^2})^2} e^{\int +(\frac{1}{x})\,dx}\,dx \\ &= \int x e^{2x^2}\,dx \\ &= \frac{1}{4}e^{2x^2}. \end{aligned}$$

Therefore $y_2(x) = \frac{1}{4}e^{2x^2} \cdot e^{x^2} = \frac{1}{4}e^{-x^2}$ *and the general solution is*

$$\begin{aligned} y(x) &= c_1 e^{-x^2} + c_2 \frac{1}{4} e^{x^2} \\ &= c_1 e^{-x^2} + c_3 e^{x^2}, \quad \textit{where } c_3 = \frac{1}{4}c_2. \end{aligned}$$

Chapter 6

Method of Variation of Parameters

Consider the nonhomogeneous linear second order differential equation

$$y''(x) + p(x)y'(x) + q(x)y(x) = g(x). \tag{6.1}$$

The general solution is comprised of the homogeneous solution $y_h(x)$ and the particular solution $y_p(x)$.

The method of variation parameter is used to find the particular solution once we have obtained the homogeneous solution,

$$y_h(x) = c_1 y_1(x) + c_2 y_2(x), \tag{6.2}$$

where all c_1 and c_2 are constants. To find the particular solution to the nonhomogeneous equation, the idea behind variation of parameter is to replace the constant in (6.2) by function of x. That is we seek a solution of (6.1) of the form

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x). \tag{6.3}$$

where the procedure to find these functions, $u_1(x)$ and $u_2(x)$ are described below.

How to determine a particular solution to $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$

(a) Find a fundamental solution set $\{y_1(x), y_2(x)\}$ for the corresponding homogeneous equation and take

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

(b) determine $u_1(x)$ and $u_2(x)$ by solving the the following expressions

$$u_1(x) = -\int \frac{g(x)\ y_2(x)}{W(y_1, y_2)}\ dx,$$
$$u_2(x) = \int \frac{g(x)\ y_1(x)}{W(y_1, y_2)}\ dx$$

where $W(y_1, y_2)$ is the Wronskian.

Example 18. Solve the following equation

$$y'' - 3y' + 2y = xe^x + 2x.$$

Solution. *The characteristic equation is* $m^2 - 3m + 2 = 0$ *which implies* $m = 1, 2$.

The homogeneous solution is

$$y_h(x) = c_1e^x + c_2e^{2x}$$

for some constants c_1 *and* c_2.

Let $y_p = u_1y_1 + u_2y_2$ *where* $y_1 = e^x$ *and* $y = e^{2x}$.

The Wronskian is

$$W[y_1, y_2](x) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x}.$$

Then

$$\begin{aligned} u_1(x) &= -\int \frac{g(x)y_2(x)}{W(y_1, y_2)}\ dx \\ &= -\int \frac{(xe^x + 2x)e^{2x}}{e^{3x}}\ dx \\ &= -\int x + 2xe^{-x}\ dx \\ &= -\frac{x^2}{2} + 2xe^{-x} + 2e^{-x}\ dx. \end{aligned}$$

and

$$\begin{aligned} u_2(x) &= \int \frac{g(x)y_1(x)}{W(y_1, y_2)}\ dx \\ &= \int \frac{(xe^x + 2x)e^x}{e^{3x}}\ dx \\ &= \int xe^{-x} + 2xe^{-2x}\ dx \\ &= -xe^{-x} - e^{-x} - xe^{-2x} - \tfrac{1}{2}e^{-2x}. \end{aligned}$$

The particular integral is

$$\begin{aligned} y_p &= \left[-\tfrac{x^2}{2} + 2xe^{-x} + 2e^{-x}\right] e^x + \left[-xe^{-x} - e^{-x} - xe^{-2x} - \tfrac{1}{2}e^{-2x}\right] e^{2x} \\ &= -\tfrac{x^2}{2} - xe^{-x} - e^{-x} + \tfrac{3}{2}. \end{aligned}$$

The general solution is then

$$y(x) = c_1 e^x + c_2 e^{2x} - \tfrac{x^2}{2} - xe^{-x} - e^{-x} + \tfrac{3}{2}.$$

Example 19. Solve the following equation

$$y'' + y = \tan x.$$

Solution. *The characteristic equation is* $m^2 + 1 = 0$ *which implies* $m = \pm i$.
The homogeneous solution is

$$y_h(x) = c_1 \cos x + c_2 \sin x.$$

Let $y_p = u_1 y_1 + u_2 y_2$ *where* $y_1 = \cos x$ *and* $y_2 = \sin x$. *The Wronskian is*

$$W[y_1, y_2](x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^x = 1.$$

Then

$$\begin{aligned} u_1(x) &= -\int \frac{g(x)y_2(x)}{W(y_1, y_2)}\, dx \\ &= -\int (\tan x)(\sin x)\, dx \\ &= -\int \frac{\sin x}{\cos x}(\sin x)\, dx \\ &= -\int \frac{\sin^2 x}{\cos x}\, dx \\ &= -\int \sec x - \cos x\, dx \\ &= -\ln|\sec x + \tan x| + \sin x, \end{aligned}$$

and

$$\begin{aligned} u_2(x) &= \int \frac{g(x)y_1(x)}{W(y_1, y_2)}\ dx \\ &= \int (\tan x)(\cos x)\ dx \\ &= \int \frac{\sin x}{\cos x}(\cos x)\ dx \\ &= \int \sin x\ dx \\ &= -\cos x. \end{aligned}$$

The particular integral is

$$y_p = (-\ln|\sec x + \tan x| + \sin x)\cos x - \cos x \sin x \quad = -\cos x \ln|\sec x + \tan x|.$$

The general solution is then

$$y(x) = c_1 \cos x + c_2 \sin x - \cos x \ln|\sec x + \tan x|.$$

Chapter 7

Power series - a short introduction

Definition 3. A power series is an infinite series about a point x_0 of the form

$$\sum_{k=0}^{\infty} a_k(x-x_0)^k = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots \tag{7.1}$$

where x is a variable and $a_0, a_1, a_2, \ldots$ are constants. The point x_0 is known as a centre point of this series.

If $x_0 = 0$, we obtain a power series in powers of x,

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \cdots$$

All variable and constant are assumed to be real.

7.1 Convergence and divergence of a power series

A power series may converge or diverge. The radius of convergence is a number R such that the series (7.1) converges absolutely for $|x-x_0| < R$, and diverges for $|x-x_0| > R$. At $|x-x_0| = R$ the series may converge or diverge.

The interval $|x-x_0| < R$ or $-R < x < R$ with possible inclusion of endpoints, is called the interval of convergence of the power series. This R is called the radius of convergence and can be found using these formulae

$$R = \frac{1}{\lim_{k\to\infty} |a_k|^{1/k}} \quad \text{or} \quad R = \frac{1}{\lim_{k\to\infty} |a_{k+1}/a_k|^{1/k}}.$$

7.2 Some familiar power series

$$
\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots, && -\infty < x < \infty \\
\cosh x &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \cdots, && -\infty < x < \infty \\
\sinh x &= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + \frac{x^{2n+1}}{(2n+1)!} + \cdots, && -\infty < x < \infty \\
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots, && -\infty < x < \infty \\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots, && -\infty < x < \infty \\
\ln|1+x| &= x - \frac{x^3}{3} + \frac{x^5}{5} + \cdots + (-1)^n \frac{x^n}{n} + \cdots, && -1 < x < 1 \\
\frac{1}{1-x} &= 1 + x + x^2 + x^3 + \cdots && -1 < x < 1
\end{aligned}
$$

7.3 Algebraic operations on power series

Let

$$y_1 = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots = \sum_{k=0}^{\infty} a_k x^k$$

and

$$y_2 = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n + \cdots = \sum_{k=0}^{\infty} b_k x^k$$

be two power series.

7.3.1 Addition and substraction between two power series

$$
\begin{aligned}
y_1 \pm y_2 &= (a_0 \pm b_0) + (a_1 \pm b_1)x + (a_2 \pm b_2)x^2 + \cdots + (a_n + b_n)x_n + \cdots \\
&= \sum_{k=0}^{\infty} (a_n + b_n) x_n.
\end{aligned}
$$

7.3.2 Multiplication between two power series

$$y_1 \cdot y_2 = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots$$

where

$$\begin{aligned} c_0 &= a_0 b_0 \\ c_1 &= a_0 b_1 + a_1 b_0 \\ c_2 &= a_0 b_2 + a_1 b_1 + a_2 b_0 \\ &\vdots \\ c_n &= a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k} \end{aligned}$$

7.3.3 Derivative of a power series

If

$$y(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \cdots = \sum_{k=0}^{\infty} a_k x^k$$

then

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots + a_n x^n + \cdots = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

7.4 Ordinary and singular Points

A homogeneous linear second order ODE can be written as

$$a(x)y'' + b(x)y' + c(x)y = 0. \tag{7.2}$$

If $a(x) \neq 0$ we can divide equation (7.2) by an $a(x)$ to obtain the standard form

$$y'' + p(x)y' + q(x)y = 0 \tag{7.3}$$

where $p(x) = \frac{b(x)}{a(x)}$ and $q(x) = \frac{c(x)}{a(x)}$.

7.4.1 Ordinary Points

A point x_0 is an ordinary point of the equation (7.3) if when x is substituted by x_0 in $p(x)$ and $q(x)$, then $p(x_0)$ and $q(x_0)$ are finite.

Example 20. The equation $y''+xy'+(x^2-4)y=0$ is in standard form with $p(x)=x$ and $q(x)=x^2-4$. All point $x_0 \in R$ are ordinary points because $p(x_0)=x_0$ and $q(x_0)=x_0^2-4$ are finite.

Example 21. Consider the equation $(x-1)y''+xy'+\frac{1}{x}y=0$. In standard form it is $y''+\frac{x}{x-1}y'+\frac{1}{x(x-1)}y=0$ where $p(x)=\frac{x}{x-1}$ and $q(x)=\frac{1}{x(x-1)}$. Here all points except 0 and 1 are ordinary points because $p(1)$ and $q(0)$ are not finite.

Example 22. The equation $x^2(x-2)^2y''+2(x-2)y'+(x+1)y=0$ in standard form is $y''+\frac{2}{x^2(x-2)}y'+\frac{x+1}{x^2(x-2)}y=0$ where $p(x)=\frac{2}{x^2(x-2)}$ and $q(x)=\frac{x+1}{x^2(x-2)}$. Here all points except 0 and 2 are ordinary points because $p(0), q(0)$ and $p(2), q(2)$ are not finite.

7.4.2 Singular Points

If $p(x_0)$ or $q(x_0)$ or both are not finite then x_0 is not an ordinary point (or called as a singular point). There are two kinds of singular points

(a) Regular singular point

(b) Irregular singular point

Regular singular point

A singular point x_0 is a regular singular point if

$$\lim_{x\to x_0}(x-x_0)p(x) \quad \text{and} \quad \lim_{x\to x_0}(x-x_0)^2q(x) \tag{7.4}$$

are both finite.

Irregular singular point

A singular point x_0 is an irregular singular point if any one or both of (7.4) are not finite.

Example 23. In Example 20, all points are ordinary points, therefore there is no singular points.

Example 24. In Example 21, the points 0 and 1 are singular points because $p(1)$ and $q(0)$ are not finite. Here, $p(x)=\frac{x}{x-1}$ and $q(x)=\frac{1}{x(x-1)}$.

For the point $x_0 = 0$,

$$\lim_{x\to 0}(x-0)p(x) = \lim_{x\to 0}\frac{x^2}{x-1} = 0,$$
$$\text{and} \quad \lim_{x\to 0}(x-0)^2 q(x) = \lim_{x\to 0}\frac{x}{x-1} = 0,$$

therefore 0 is a regular singular point.

For the point $x_0 = 1$,

$$\lim_{x\to 1}(x-1)p(x) = \lim_{x\to 1} x = 1,$$
$$\text{and} \quad \lim_{x\to 1}(x-1)^2 q(x) = \lim_{x\to 1}\frac{x-1}{x} = 0,$$

wherefore 1 is a regular singular point.

Example 25. In Example 22, the points 0 and 2 are singular points because $p(0), q(0)$ and $p(2), q(2)$ are not finite. Here, $p(x) = \frac{2}{x^2(x-2)}$ and $q(x) = \frac{x+1}{x^2(x-2)^2}$.

For the point $x_0 = 0$,

$$\lim_{x\to 0}(x-0)p(x) = \lim_{x\to 0}\frac{2}{x(x-2)} = \infty,$$
$$\text{and} \quad \lim_{x\to 0}(x-0)^2 q(x) = \lim_{x\to 0}\frac{x+1}{(x-2)^2} = \tfrac{1}{4},$$

therefore 0 is an irregular singular point.

For the point $x_0 = 2$,

$$\lim_{x\to 2}(x-2)p(x) = \lim_{x\to 2}\frac{2}{x^2} = \tfrac{1}{2}$$
$$\text{and} \quad \lim_{x\to 2}(x-2)^2 q(x) = \lim_{x\to 2}\frac{x+1}{x^2} = \tfrac{3}{4},$$

therefore 2 is a regular singular point.

Chapter 8

Series Solutions of ODEs

Consider a general second order linear homogeneous differential equations

$$p(x)y''(x) + q(x)y'(x) + r(x)y(x) = 0 \tag{8.1}$$

where $p(x) \neq 0$ on the interval $\alpha < x < \beta$. We knew that every solution to the equation (8.1) can be written in the form

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where $y_1(x)$ and $y_2(x)$ are two linearly independent solutions of (8.1). If $p(x), q(x)$, and $r(x)$ are constants, we have a method of obtaining these solutions.

Suppose $p(x), q(x)$, and $r(x)$ are polynomials in x, then the format of the equation (8.1) suggest that $y(x)$ should be a polynomial too. If $y(x)$ is a polynomial, then $p(x)y'', q(x)y'(x)$ and $r(x)y(x)$ are all polynomials in x.

In principal, we can determine the coefficients of $y(x)$, a solution to (8.1), by equating the summation of coefficients of the same power of x on the left hand side of equation (8.1) to be zero since the right hand side of (8.1) can be equated to a polynomial of zero coefficients.

Example 26. Find two linearly independent solutions to the equation

$$y'' - 2xy' - 2y = 0. \tag{8.2}$$

Solution. *Since we suppose that the solution to* (8.2) *is a polynomial, but we do not have any idea of its highest term, we let*

$$y(x) = a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{k=0}^{\infty} a_k x^k \tag{8.3}$$

Taking the derivative, we have

$$y'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots = \sum_{k=1}^{\infty} ka_kx^{k-1} \tag{8.4}$$

and

$$y''(x) = 2a_2 + 6a_3x + 12a_4x^2 + \cdots = \sum_{k=2}^{\infty} k(k-1)a_kx^{k-2}. \tag{8.5}$$

Substituting $y(x)$ and its derivatives into equation (8.2), we obtain

$$\begin{gathered}\left[\sum_{k=2}^{\infty} k(k-1)a_kx^{k-2}\right] - 2x\left[\sum_{k=1}^{\infty} ka_kx^{k-1}\right] - 2\left[\sum_{k=0}^{\infty} a_kx^k\right] = 0 \\ \left[\sum_{k=2}^{\infty} k(k-1)a_kx^{k-2}\right] - 2\left[\sum_{k=1}^{\infty} ka_kx^k\right] - 2\left[\sum_{k=0}^{\infty} a_kx^k\right] = 0\end{gathered} \tag{8.6}$$

The next step is to change the summation in (8.6) so that all the exponential terms are k, not $k-2$, or vice versa. This can be achieved by the raising each value of k under the first summation by 2 and reducing

This results in all the coefficients of y to be uniquely determined by a_0 and a_1. The values of a_0 and a_1 are arbitrary. Since

$$y(x) = a_0 + a_1x + a_2x^2 + \cdots,$$

the values of y and y' at $x = 0$ are a_0 and a_1 respectively. The values a_0 and a_1 are uniquely determined when we impose an initial value condition on y and y'. To obtain two linearly independent solutions, notice that all the even coefficients namely $a_2, a_4, a_6, \ldots$ are reduced to some constant multiple of a_0, while the odds coefficiens, $a_3, a_5, a_7, \ldots$ are reduced to some constant multiple of a_1 .

Here we have two choice either replace k by $k+2$ in the first summation so that all exponential terms are k or replace k by $k-2$ in the second and third summation.

8.0.1 First choice, all exponential terms are k

From

$$\left[\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}\right] - 2\left[\sum_{k=1}^{\infty} k a_k x^k\right] - 2\left[\sum_{k=0}^{\infty} a_k x^k\right] = 0$$

we replace k by $k+2$ in the first summation.

$$\left[\sum_{k+2=2}^{\infty} (k+2)([k+2]-1)a_{k+2}x^{(k+2)-2}\right] - 2\left[\sum_{k=1}^{\infty} k a_k x^k\right] - 2\left[\sum_{k=0}^{\infty} a_k x^k\right] = 0$$

$$\left[\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k\right] - 2\left[\sum_{k=1}^{\infty} k a_k x^k\right] - 2\left[\sum_{k=0}^{\infty} a_k x^k\right] = 0$$

$$2\cdot 1\cdot a_2 + \left[\sum_{k=1}^{\infty} (k+2)(k+1)a_{k+2}x^k\right] - 2\left[\sum_{k=1}^{\infty} k a_k x^k\right] - 2a_0 - 2\left[\sum_{k=1}^{\infty} a_k x^k\right] = 0$$

$$2a_2 - 2a_0 + \sum_{k=1}^{\infty}\Big[(k+2)(k+1)a_{k+2} - 2k a_k - 2a_k\Big]x^k = 0$$

This implies

$$2a_2 - 2a_0 = 0, \quad \textit{and} \quad (k+2)(k+1)a_{k+2} - 2ka_k - 2a_k = 0$$

or

$$a_2 = a_0, \quad \textit{and} \quad (k+2)(k+1)a_{k+2} - 2(k+1)a_k = 0$$

or

$$a_2 = a_0, \quad \textit{and} \qquad a_{k+2} = \frac{2(k+1)}{(k+2)(k+1)}a_k$$

$$a_{k+2} = \frac{2}{k+2}a_k, \quad k \geq 1.$$

8.0.2 Second choice, all exponential terms are $k-2$

From

$$\left[\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}\right] - 2\left[\sum_{k=1}^{\infty} k a_k x^{k}\right] - 2\left[\sum_{k=0}^{\infty} a_k x^{k}\right] = 0$$

replace k by $k-2$ in the second and third summation.

$$\left[\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}\right] - 2\left[\sum_{k-2=1}^{\infty} (k-2)a_{k-2} x^{k-2}\right] - 2\left[\sum_{k-2=0}^{\infty} a_{k-2} x^{k-2}\right] = 0$$

$$\left[\sum_{k=2}^{\infty} k(k-1)a_k x^{k-2}\right] - 2\left[\sum_{k=3}^{\infty} (k-2)a_{k-2} x^{k-2}\right] - 2\left[\sum_{k=2}^{\infty} a_{k-2} x^{k-2}\right] = 0$$

$$2\cdot 1\cdot a_2 + \left[\sum_{k=3}^{\infty} k(k-1)a_k x^{k-2}\right] - 2\left[\sum_{k=3}^{\infty} (k-2)a_{k-2} x^{k-2}\right] - 2a_0 - 2\left[\sum_{k=3}^{\infty} a_{k-2} x^{k-2}\right] = 0$$

$$2a_2 - 2a_0 + \sum_{k=3}^{\infty}\Big[k(k-1)a_k - 2(k-2)a_{k-2} - 2a_{k-2}\Big]x^{k-2} = 0$$

$$2a_2 - 2a_0 + \sum_{k=3}^{\infty}\Big[k(k-1)a_k - \big[2(k-2)+2\big]a_{k-2}\Big]x^{k-2} = 0$$

$$2a_2 - 2a_0 + \sum_{k=3}^{\infty}\Big[k(k-1)a_k - (2k-2)a_{k-2}\Big]x^{k-2} = 0$$

$$2a_2 - 2a_0 + \sum_{k=3}^{\infty}\Big[k(k-1)a_k - 2(k-1)a_{k-2}\Big]x^{k-2} = 0$$

This implies

$$2a_2 - 2a_0, = 0 \qquad \textit{and} \qquad k(k-1)a_k - 2(k-1)a_{k-2} = 0$$

or

$$a_2 = a_0, \quad \textit{and} \qquad a_k = \frac{2(k-1)}{k(k-1)}a_{k-2}$$

or

$$a_2 = a_0, \quad \textit{and} \qquad a_k = \frac{2}{k}a_{k-2}, \quad k \geq 3.$$

In the first choice we obtain the equation $a_{k+2} = \frac{2}{k+2}a_k,\ k \geq 1$ *and in the second choice we obtain the equation* $a_k = \frac{2}{k}a_{k-2},\ k \geq 3$. *These equations are known as the recurrence formula for the coefficients* $a_0, a_1, a_2, \ldots$ *For the first choice the value of* a_k *determines the value of* a_{k+2} *and for the second choice the value of* a_{k-2} *determines the value of* a_k.

In the first choice,

$$a_3 = \tfrac{2}{3}a_1 \qquad a_4 = \tfrac{2}{4}a_2 = \tfrac{2}{4}a_0$$

$$a_5 = \tfrac{2}{5}a_3 = \tfrac{2\cdot 2}{3\cdot 5}a_1 \qquad a_6 = \tfrac{2}{6}a_4 = \tfrac{2\cdot 2}{4\cdot 6}a_0$$

$$a_7 = \tfrac{2}{7}a_5 = \tfrac{2\cdot 2\cdot 2}{3\cdot 5\cdot 7}a_1 \qquad a_8 = \tfrac{2}{8}a_6 = \tfrac{2\cdot 2\cdot 2}{4\cdot 6\cdot 8}a_0$$

$$\vdots$$

In the second choice,

$$a_3 = \tfrac{2}{3}a_1 \qquad a_4 = \tfrac{2}{4}a_2 = \tfrac{2}{4}a_0$$

$$a_5 = \tfrac{2}{5}a_3 = \tfrac{2\cdot 2}{3\cdot 5}a_1 \qquad a_6 = \tfrac{2}{6}a_4 = \tfrac{2\cdot 2}{4\cdot 6}a_0$$

$$a_7 = \tfrac{2}{7}a_5 = \tfrac{2\cdot 2\cdot 2}{3\cdot 5\cdot 7}a_1 \qquad a_8 = \tfrac{2}{8}a_6 = \tfrac{2\cdot 2\cdot 2}{4\cdot 6\cdot 8}a_0$$

$$\vdots$$

It does not matter which choices we make, we end up with the same answer.

By induction, the even coefficients are

$$\begin{aligned} a_{2k} &= \frac{2^k}{4\cdot 6\cdot 8\cdots 2k}a_0 \\ &= \frac{2^k}{(2\cdot 2)\cdot(2\cdot 3)\cdot(2\cdot 4)\cdots 2k}a_0 \\ &= \frac{1}{2\cdot 3\cdot 4\cdots k}a_0 \\ &= \frac{1}{k!}a_0, \quad k \geq 1, \end{aligned}$$

and the odd coefficients are

$$a_{2k+1} = \frac{2^k}{3 \cdot 5 \cdot 7 \cdots (2k+1)} a_1.$$

Therefore the solutions to (8.2) are:

- *given by the even coefficients:*

$$\begin{aligned} y_1(x) &= a_0 + a_2 x^2 + a_4 x^4 + \cdots + a_{2k} x^{2k} + \cdots \\ &= a_0 + a_0 x^2 + \frac{1}{2} a_0 x^4 + \cdots + \frac{1}{k!} a_0 x^{2k} + \cdots \\ &= a_0 \sum_{m=0}^{\infty} \frac{1}{k!} x^{2m} \\ &= e^{x^2}, \end{aligned}$$

- *given by the odd coefficients:*

$$\begin{aligned} y_2(x) &= a_1 + a_3 x^3 + a_5 x^5 + \cdots + a_{2k+1} x^{2k+1} + \cdots \\ &= a_1 \sum_{m=0}^{\infty} \frac{2^m}{3 \cdot 5 \cdot 7 \cdots (2m+1)} x^{2m+1}. \end{aligned}$$

From the above example, we found that $y_1(x)$ and $y_2(x)$ are polynomials of infinite degree even though $p(x) = 1, q(x) = -2x$, and $r(x) = -2$ are polynomials of finite degree. Such polynomials are known as power series.

8.1 The idea behind the power series method

For a given equation

$$y'' + p(x)y' + q(x)y = 0 \tag{8.7}$$

where $p(x)$ and $q(x)$ are polynomials. We assume a solution in the form of a power series with unkwown coefficients,

$$y(x) = a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{k=0}^{\infty} a_k x^k \tag{8.8}$$

and insert this series and the series obtained by termwise differentiation,

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

$$y''(x) = 2a_2 + 6a_3 x + 12a_4 x^2 + \cdots = \sum_{k=2}^{\infty} k(k-1) a_k$$

into equation (8.7). Then we collect like powers of x and equate the sum of coefficients of each occuring power of x to zero, starting with the constant terms, the terms containing x, the terms containg x^2, etc. This gives relations from which we can determine the unkown coefficients in (8.8) succesfully.

Chapter 9

The Frobenius method

Let

$$x^2y'' + xP(x)y' + Q(x)y = 0 \tag{9.1}$$

and in its standard form

$$y'' + \frac{P(x)}{x}y' + \frac{Q(x)}{x^2}y = 0 \tag{9.2}$$

with $\lim_{x\to 0} x\frac{P(x)}{x}$ and $\lim_{x\to 0} x^2\frac{Q(x)}{x^2}$ be finite, so that 0 is a regular singular point of the equation (9.1).

The method of Frobenius says if 0 is a regular singular point of equation (9.1) then it has a series solution of the form

$$.y(x) = x^r \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+r}$$

where r satisfies a quadratic equation called the indicial equation.

Since r is a root of a quadratic equation, there are three cases to be considered and the solutions of (9.1) depends on these roots.

Case 1: Distinct roots not differing by an integer

Suppose the roots are r_1, r_2 with $r_1 > r_2$ and $r_1 - r_2 \notin \mathbf{Z}$. Then the two linearly independent solutions are

$$y_1(x) = x^{r_1}(a_0 + a_1x + a_2x_2 + \cdots)$$
$$y_2(x) = x^{r_2}(b_0 + b_1x + b_2x_2 + \cdots)$$

Case 2: Double root $r_1 = r_2 = r$

The linearly independent solutions are

$$y_1(x) = x^r(a_0 + a_1x + a_2x_2 + \cdots)$$
$$y_2(x) = y_1(x)\ln x + x^r(b_1x + b_2x_2 + \cdots)$$

Note: The second solution $y_2(x)$ may also be otained by using the reduction of order method.

Case 3: Roots differing by an integer

Suppose the roots are r_1, r_2 with $r_1 > r_2$ and $r_1 - r_2 \in \mathbb{Z}$. The the two linearly independent solutions are

$$y_1(x) = x^{r_1}(a_0 + a_1x + a_2x_2 + \cdots)$$
$$y_2(x) = y_1(x)\ln x + x^{r_2}(b_0 + b_1x + b_2x_2 + \cdots)$$

Example 27. Case 1: Distinct roots not differing by an integer.
Solve the equation

$$4xy'' + 2y' + y = 0. \tag{9.3}$$

Solution. *In the form* (9.1) *the equation is*

$$x^2y'' + \frac{1}{2}xy' + \frac{1}{4}xy = 0.$$

with $P(x) = \frac{1}{2}$ *and* $Q(x) = \frac{x}{4}$ *and in standard form* (9.2)

$$y'' + \frac{1/2}{x}y' + \frac{1/4}{x}y = 0.$$

with $p(x) = \frac{1/2}{x}$ *and* $q(x) = \frac{1/4}{x}$.

The point 0 is a singular point because $p(0) = \frac{1/2}{0}$ *and* $q(0) = \frac{1/4}{0}$ *are not finite.*

$$\lim_{x\to 0}(x-0)p(x) = \lim_{x\to 0}(x-0)\cdot\frac{1/2}{x} = \lim_{x\to 0}\frac{1}{2} = \frac{1}{2},$$

and $$\lim_{x\to 0}(x-0)^2q(x) = \lim_{x\to 0}(x)^2\cdot\frac{1/4}{x} = \lim_{x\to 0}\frac{1}{4}x = 0.$$

Hence, 0 is the regular singular point. By the method of Frobenius, it has a series solution of the form $y = \sum_{k=0}^{\infty} a_k x^{k+r}$. *Then*

$$y' = \sum_{k=0}^{\infty}(k+r)a_k x^{k+r-1}$$

$$y'' = \sum_{k=0}^{\infty}(k+r)(k+r-1)a_k x^{k+r-2}$$

Substitute into equation (9.3),

$$4x\left[\sum_{k=0}^{\infty}(k+r)(k+r-1)a_k x^{k+r-2}\right] + 2\left[\sum_{k=0}^{\infty}(k+r)a_k x^{k+r-1}\right] + \sum_{k=0}^{\infty} a_k x^{k+r} = 0$$

$$\left[\sum_{k=0}^{\infty} 4(k+r)(k+r-1)a_k x^{k+r-1}\right] + \left[\sum_{k=0}^{\infty} 2(k+r)a_k x^{k+r-1}\right] + \sum_{k=0}^{\infty} a_k x^{k+r} = 0$$

Make the power of x *in each summation to be the same by substituting* k *with* $k-1$ *in the third summation*

$$4x\left[\sum_{k=0}^{\infty}(k+r)(k+r-1)a_k x^{k+r-2}\right] + 2\left[\sum_{k=0}^{\infty}(k+r)a_k x^{k+r-1}\right] + \sum_{k-1=0}^{\infty} a_k x^{(k-1)+r} = 0$$

$$\left[\sum_{k=0}^{\infty} 4(k+r)(k+r-1)a_k x^{k+r-1}\right] + \left[\sum_{k=0}^{\infty} 2(k+r)a_k x^{k+r-1}\right] + \sum_{k=1}^{\infty} a_{k-1} x^{k+r-1} = 0$$

Make the index of each summation starts at the same number $k = 1$

$$4r(r-1)a_0 x^{r-1} + \left[\sum_{k=1}^{\infty} 4(k+r)(k+r-1)a_k x^{k+r-1}\right]$$
$$+2ra_0 x^{r-1} + \left[\sum_{k=1}^{\infty} 2(k+r)a_k x^{k+r-1}\right]$$
$$+\sum_{k=1}^{\infty} a_{k-1} x^{k+r-1} = 0$$

Simplify and express as a single summation,

$$4r(r-1)a_0 x^{r-1} + 2ra_0 x^{r-1}$$
$$+\sum_{k=1}^{\infty}\Big[4(k+r)(k+r-1)a_k + 2(k+r)a_k + a_{k-1}\Big]x^{k+r-1} = 0$$

factorizing,

$$\Big[4r(r-1)+2r\Big]a_0x^{r-1}+\sum_{k=1}^{\infty}\Big[(k+r)\big[4(k+r-1)+2\big]a_k+a_{k-1}\Big]x^{k+r-1}=0$$

$$\Big[4r(r-1)+2r\Big]a_0x^{r-1}+\sum_{k=1}^{\infty}\Big[(k+r)\big(4k+4r-4+2\big)a_k+a_{k-1}\Big]x^{k+r-1}=0$$

and simplifying,

$$\Big[4r(r-1)+2r\Big]a_0x^{r-1}+\sum_{k=1}^{\infty}\Big[(k+r)\big(4k+4r-2\big)a_k+a_{k-1}\Big]x^{k+r-1}=0$$

$$\Big[4r(r-1)+2r\Big]a_0x^{r-1}+\sum_{k=1}^{\infty}\Big[2(k+r)(2k+2r-1)a_k+a_{k-1}\Big]x^{k+r-1}=0$$

This implies

$$\begin{aligned}\big[4r(r-1)+2r\big]a_0&=0\\(4r^2-2r)a_0&=0,\end{aligned}$$

and

$$2(k+r)(2k+2r-1)a_k+a_{k-1}=0.$$

Since by definition $a_0\neq 0$, then

$$\begin{aligned}4r^2-2r&=0,\quad \textit{this is the indicial equation}\\r&=0,\tfrac{1}{2}.\end{aligned}$$

If we let $r_1=\frac{1}{2}$ and $r_2=0$, then $r_1-r_2=\frac{1}{2}$ which is not integer [or if we let $r_1=0$ and $r_2=\frac{1}{2}$, then $r_1-r_2=-\frac{1}{2}$ which is not integer.]

The recurrence relation is

$$2(k+r)(2k+2r-1)a_k+a_{k-1}=0$$

$$a_k=\frac{-1}{2(k+r)(2k+2r-1)}a_{k-1},\quad k=1,2,3,\ldots$$

If $r=\frac{1}{2}$, then

$$a_k=\frac{-1}{2(k+\frac{1}{2})(2k+2\cdot\frac{1}{2}-1)}a_{k-1}=\frac{-1}{(2k+1)(2k+1-1)}a_{k-1}=\frac{-1}{2k(2k+1)}a_{k-1},$$

for

$$\begin{aligned} k=1, \qquad & a_1 = \frac{-1}{2\cdot 3}a_0 = -\frac{a_0}{3!} \\ k=2, \qquad & a_2 = \frac{-1}{4\cdot 5}a_1 = \frac{-1}{4\cdot 5}\cdot -\frac{a_0}{3!} = \frac{a_0}{5!} \\ k=3, \qquad & a_3 = \frac{-1}{6\cdot 7}a_2 = \frac{-1}{6\cdot 7}\cdot \frac{a_0}{5!} = -\frac{a_0}{7!} \\ & \vdots \end{aligned}$$

In general

$$a_k = \frac{(-1)^k}{(2k+1)!}\ a_0.$$

Therefore a linearly independent solution is

$$\begin{aligned} y_1(x) &= x^{1/2}(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + \cdots) \\ &= x^{1/2}\left(a_0 - \frac{x}{3!} + \frac{x^2}{5!} - \frac{x^3}{7!} + - \cdots + \frac{(-1)^n}{(2n+1)!}x^n + \cdots\right) \end{aligned}$$

let $a_0 = 1$,

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}x^{k+1/2}.$$

If $r = 0$, *then*

$$a_k = \frac{-1}{2(k+0)(2k+2\cdot 0-1)}a_{k-1} = \frac{-1}{2k(2k+0-1)}a_{k-1} = \frac{-1}{2k(2k-1)}a_{k-1},$$

for

$$\begin{aligned} k=1, \qquad & a_1 = \frac{-1}{2\cdot 1}a_0 = -\frac{a_0}{2!} \\ k=2, \qquad & a_2 = \frac{-1}{4\cdot 3}a_1 = \frac{-1}{4\cdot 3}\cdot -\frac{a_0}{2!} = \frac{a_0}{4!} \\ k=3, \qquad & a_3 = \frac{-1}{5\cdot 6}a_2 = \frac{-1}{5\cdot 6}\cdot \frac{a_0}{4!} = -\frac{a_0}{6!} \\ & \vdots \end{aligned}$$

In general

$$a_k = \frac{(-1)^k}{(2k)!}\ a_0.$$

Therefore another linearly independent solution is

$$\begin{aligned} y_2(x) &= x^0(b_0 + b_1x + b_2x^2 + \cdots + b_nx^n + \cdots) \\ &= 1\left(b_0 - \frac{x}{2!} + \frac{x^2}{4!} - \frac{x^3}{6!} + - \cdots + \frac{(-1)^n}{(2n)!}x^n + \cdots\right) \end{aligned}$$

let $b_0 = 1$,

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}x^k.$$

The general solution is then

$$y(x) = Ay_1(x) + By_2(x).$$

Example 28. Case 3: Distinct roots not differing by an integer
Solve the Bessel's equation $x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0$.

Solution. *Here* $P(x) = 1$ *and* $Q(x) = x^2 - \frac{1}{4}$.
In standard form

$$y'' + \frac{1}{x}y' + \left(1 - \frac{1}{4x^2}\right)y = 0.$$

with $p(x) = \frac{1}{x}$ *and* $q(x) = 1 - \frac{1}{4x^2}$.
Now $x = 0$ *is the singular point, and*

$$\begin{aligned} &\lim_{x\to 0} xp(x) = \lim_{x\to 0} x \cdot \frac{1}{x} = \lim_{x\to 0} 1 = 1, \\ \textit{and} \quad &\lim_{x\to 0} x^2q(x) = \lim_{x\to 0} x^2\left(1 - \frac{1}{4x^2}\right) = \lim_{x\to 0}\left(x^2 - \frac{1}{4}\right) = -\tfrac{1}{4}, \end{aligned}$$

are both finite. Hence, $x = 0$ *is the regular singular point. The method of Frobenius say the equation has a series solution of the* $y = \sum_{k=0}^{\infty} a_kx^{k+r}$. *Then*

$$\begin{aligned} y' &= \sum_{k=0}^{\infty}(k+r)a_kx^{k+r-1} \\ y'' &= \sum_{k=0}^{\infty}(k+r)(k+r-1)a_kx^{k+r-2} \end{aligned}$$

Substitute into the equation

$$x^2\left[\sum_{k=0}^{\infty}(k+r)(k+r-1)a_kx^{k+r-2}\right]+x\left[\sum_{k=0}^{\infty}(k+r)a_kx^{k+r-1}\right]$$
$$+\left(x^2-\frac{1}{4}\right)\sum_{k=0}^{\infty}a_kx^{k+r}=0$$
$$\left[\sum_{k=0}^{\infty}(k+r)(k+r-1)a_kx^{k+r}\right]+\left[\sum_{k=0}^{\infty}(k+r)a_kx^{k+r}\right]$$
$$+\sum_{k=0}^{\infty}a_kx^{k+r+2}-\sum_{k=0}^{\infty}\frac{1}{4}a_kx^{k+r}=0$$

Make the power of x in the third summation to be x^{k+r} by substituting k by $k-2$,

$$\left[\sum_{k=0}^{\infty}(k+r)(k+r-1)a_kx^{k+r}\right]+\left[\sum_{k=0}^{\infty}(k+r)a_kx^{k+r}\right]$$
$$+\sum_{k-2=0}^{\infty}a_{k-2}x^{(k-2)+r+2}-\sum_{k=0}^{\infty}\frac{1}{4}a_kx^{k+r}=0$$
$$\left[\sum_{k=0}^{\infty}(k+r)(k+r-1)a_kx^{k+r}\right]+\left[\sum_{k=0}^{\infty}(k+r)a_kx^{k+r}\right]$$
$$+\sum_{k=2}^{\infty}a_{k-2}x^{k+r}-\sum_{k=0}^{\infty}\frac{1}{4}a_kx^{k+r}=0$$

Then

$$r(r-1)a_0x^r + (1+r)ra_1x^{1+r} + \left[\sum_{k=2}^{\infty}(k+r)(k+r-1)a_kx^{k+r}\right]$$
$$+ra_0x^r + (1+r)a_1x^{1+r} + \left[\sum_{k=2}^{\infty}(k+r)a_kx^{k+r}\right]$$
$$+\sum_{k=2}^{\infty}a_{k-2}x^{k+r}$$
$$-\frac{1}{4}a_0x^r - \frac{1}{4}a_1x^{1+r} - \sum_{k=2}^{\infty}\frac{1}{4}a_kx^{k+r} = 0$$

Collecting like terms

$$r(r-1)a_0x^r + ra_0x^r - \tfrac{1}{4}a_0x^r$$
$$+ (1+r)ra_1x^{1+r} + (1+r)a_1x^{1+r} - \tfrac{1}{4}a_1x^{1+r}$$
$$+\sum_{k=2}^{\infty}\left[(k+r)(k+r-1)a_kx^{k+r} + (k+r)a_kx^{k+r} + a_{k-2}x^{k+r} - \tfrac{1}{4}a_kx^{k+r}\right] = 0$$

and simplifying

$$\left(r^2 - r + r - \tfrac{1}{4}\right)a_0x^r + \left(r + r^2 + 1 + r - \tfrac{1}{4}\right)a_1x^{1+r}$$
$$+\sum_{k=2}^{\infty}\left\{\ \left[(k+r)(k+r-1) + (k+r) - \tfrac{1}{4}\right]a_k + a_{k-2}\ \right\}x^{k+r} = 0$$

$$(r^2 - \tfrac{1}{4})a_0x^r + \left(r^2 + 2r + 1 - \tfrac{1}{4}\right)a_1x^{1+r}$$
$$+\sum_{k=2}^{\infty}\left[\left[(k+r)\left\{(k+r-1)+1\right\} - \tfrac{1}{4}\right]a_k + a_{k-2}\right]x^{k+r} = 0$$

$$(r^2 - \tfrac{1}{4})a_0x^r + \left[(r+1)^2 - \tfrac{1}{4}\right]a_1x^{1+r}$$
$$+\sum_{k=2}^{\infty}\left[\left[(k+r)^2 - \tfrac{1}{4}\right]a_k + a_{k-2}\right]x^{k+r} = 0$$

This implies $(r^2 - \frac{1}{4})a_0 = 0$ *and since* $a_0 \neq 0$, *indicial equation* $r^2 - \frac{1}{4} = 0$ *which implies* $r_1 = \frac{1}{2}$ *and* $r_2 = -\frac{1}{2}$.

This also implies $\left[(r+1)^2 - \frac{1}{4}\right]a_1 = 0$ *and since* $r_1 = \frac{1}{2}, r_2 = -\frac{1}{2}$ *we have* $\left[(r+1)^2 - \frac{1}{4}\right]a_1 \neq 0$. *This forces* $a_1 = 0$.

Also

$$\begin{aligned}
\left[(k+r)^2 - \tfrac{1}{4}\right]a_k + a_{k-2} &= 0 \\
\left(k+r-\tfrac{1}{2}\right)\left(k+r+\tfrac{1}{2}\right)a_k &= -a_{k-2} \\
a_k &= \frac{-1}{(k+r-\frac{1}{2})(k+r+\frac{1}{2})}a_{k-2} \quad k = 2,3,4,\ldots
\end{aligned}$$

When $r_1 = \frac{1}{2}$, *the recurrence relation is*

$$a_k = \frac{-1}{(k+\frac{1}{2}-\frac{1}{2})(k+\frac{1}{2}+\frac{1}{2})}a_{k-2} = \frac{-1}{k(k+1)}a_{k-2}, \quad k = 2,3,4$$

For

$$\begin{aligned}
k = 2, &\qquad a_2 = \frac{-1}{2\cdot 3}a_0 \\
k = 3, &\qquad a_3 = \frac{-1}{3\cdot 4}a_1 = 0 \\
k = 4, &\qquad a_4 = \frac{-1}{4\cdot 5}a_2 = \frac{-1}{4\cdot 5}\cdot\frac{-1}{2\cdot 3}a_0 = \frac{1}{2\cdot 3\cdot 4\cdot 5}a_0 = \frac{1}{5!}a_0
\end{aligned}$$

The first linearly independent solution is

$$\begin{aligned}
y_1(x) &= x^{1/2}(a_0 + a_1x + a_2x^2 + a^3x^3 + a^4x^4 + \cdots) \\
&= x^{1/2}\left(a_0 - \frac{a_0}{3!}x^2 + \frac{a_0}{5!}x^4 - \frac{a_0}{7!}x^6 + \cdots\right) \\
&= a_0x^{-1/2}\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots\right) \\
&= a_0x^{-1/2}\sin x.
\end{aligned}$$

The second solution is

$$y_2(x) = Ky_1(x)\ln x + x^{-1/2}(A_0 + A_1x + A_2x^2 + \cdots)$$

and the constants $K, A_0, A_2, A_3, \ldots$ *may be determined by substituting* $y_2(x)$ *into the differential equation. This is tedious work. :)*

Chapter 10

Bessel's and Legendre's equations

The two equations

$$x^2y'' + xy' + (x^2 - v^2)y = 0 \tag{10.1}$$

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0 \tag{10.2}$$

occur frequently in advanced studies in applied mathematics, physics and engineering. They are called **Bessel's equation** and **Legendre's equation** respectively. In solving (10.1) we shall assume $v \geq 0$, whereas in (10.2) we shall consider only the case when n is a nonnegative integer.

Refer Handout

Chapter 11

Application of 2nd ODEs

Example 29. In the study of a vibrating spring with damping, we are led to an initial value problem of the form

$$mx''(t) + bx'(t) + kx(t) = 0, \quad x(0) = x_0,\ x'(0) = v_0 \tag{11.1}$$

where

m	:	the mass of the spring system,
b	:	the damping constant,
k	:	the spring constant,
x_0	:	the initial displacement,
v_0	:	is the initial velocity,
$x(t)$	:	the displacement from equilibrium of the spring system at time t.

Determine the equation of motion for this spring system when $m = 36$ kg, $b = 12$ kg sec^{-1}, $k = 37$ kg sec^{-2}, $x_0 = 70$ cm, and $v_0 = 10$ cm sec^{-1}. Also find $x(10)$, the displacement after 10 sec.

Solution. *The equation of motion is given by $x(t)$, the solution of the initial value problem (11.1) for the specified values of m, b, k, x_0 and v_0. That is we seek the solution to*

$$36x'' + 12x' + 37x = 0, \quad x(0) = 70,\ x'(0) = 10 \tag{11.2}$$

The auxiliary equation for (11.2) is $36r^2 + 12r + 37 = 0$ which has roots

$$r = \frac{-12 \pm \sqrt{144 - 4 \cdot 36 \cdot 37}}{72} = \frac{-12 \pm \sqrt{1 - 37}}{72} = -\frac{1}{6} \pm i.$$

Hence with $\alpha = \frac{1}{6}, \beta = 1$, the displacement $x(t)$ can be expressed in the form

$$x(t) = c_1 e^{-t/6} \cos t + c_2 e^{-t/6} \sin t.$$

Then

$$x'(t) = \left(-\frac{c_1}{6} + c_2\right) e^{-t/6} \cos t + \left(-c_1 - \frac{c_2}{6}\right) e^{-t/6} \sin t.$$

The initial condition $x(0) = 70$ *implies* $c_1 = 70$ *and the condition* $x'(0) = 10$ *implies* $-\frac{c_1}{6} + c_2 = 10$, *so that* $c_2 = \frac{65}{3}$.

Therefore

$$x(t) = 70e^{-t/6} \cos t + \tfrac{65}{3} e^{-t/6} \sin t,$$

and so

$$x(10) = 70e^{-t/6} \cos 10 + \tfrac{65}{3} e^{-t/6} \sin 10 \approx 13.32 \ \textit{cm}.$$

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